GradeS: Geometry-Aware Reliability-Guided Diffusion Supervision for Sparse-View 3D Gaussian Splatting

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CS599 Resource Efficient AI

Background

- Sparse-view 3DGS aims to reduce the number of required input views.
- However, even sparse-view methods …
  - often rely on moderately sparse but sufficiently overlapping views
  - focus on constrained indoor / object-centric scenes.

GT S3 S6 S9

Sparse input views provide limited coverage of the scene

- Real-world unbounded outdoor scenes are often sparsely captured and largely unobserved.
- Unbounded scenes contain both near foreground and distant background

⇒ This near/far imbalance causes unstable geometry, near foreground floaters and background artifacts

Motivation

- **Problem 1: Unstable Geometry**
  - Sparse-view 3DGS suffers from floaters and unstable geometry
- **Problem 2: Unreliable Diffusion Supervision**
  - Generated pseudo-views may be inconsistent or hallucinated
- **Key Question:**

How can we use generated views as reliable supervision for sparse-view 3DGS?

Unstable geometry

Hallucinated generated frames

- **Solution: GradeS**
  1) **Geometry-aware scene grounding before generation**
  +
  2) **Reliability-guided supervision after generation**

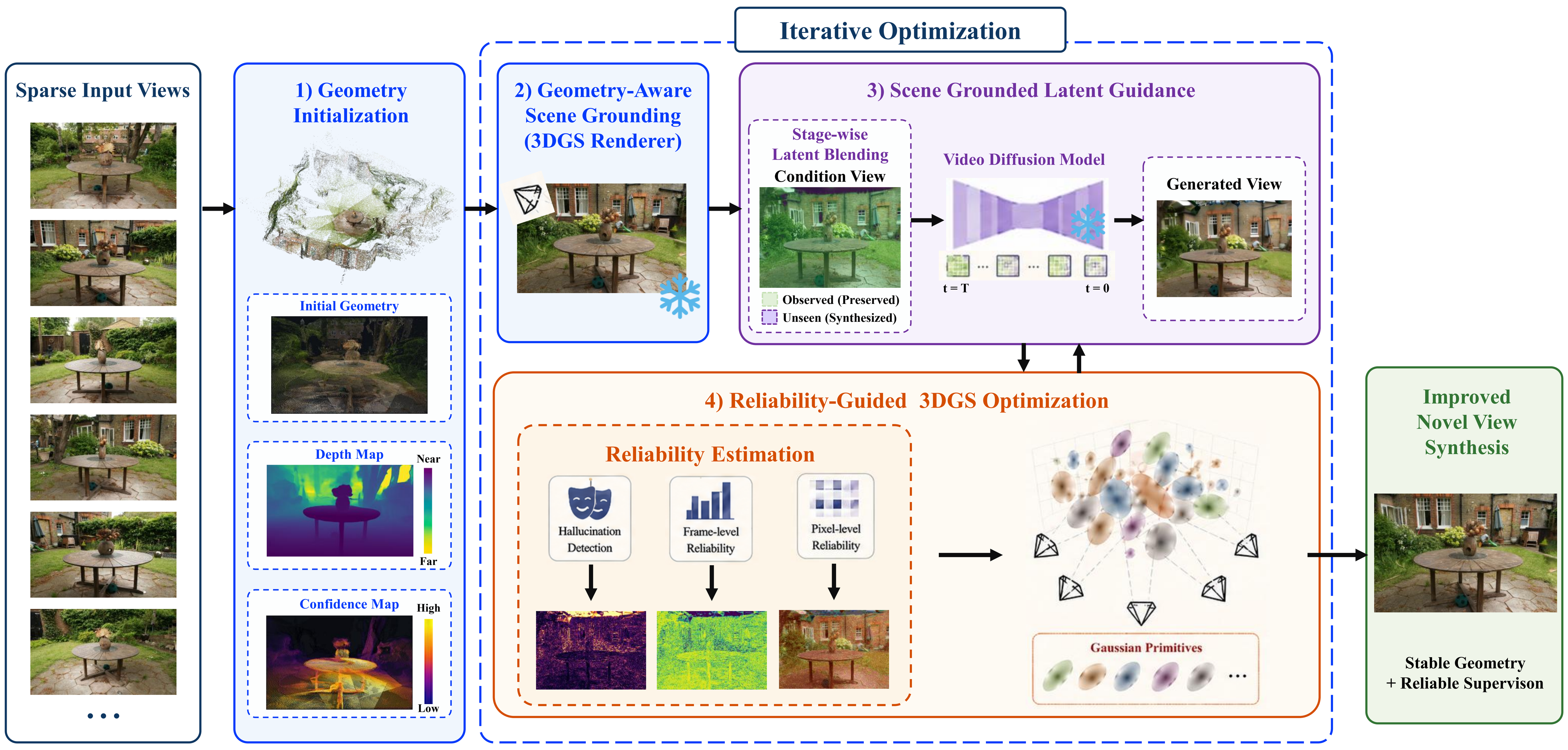
Methods

1) Geometry Initialization
- Initialize 3D Gaussians from point clouds

2) Geometry-Aware Scene Grounding
- Render baseline 3DGS RGB, depth, and confidence map

3) Scene-Grounded Latent Guidance
- Preserve observed regions via latent blending
- holes be synthesized by video diffusion

4) Reliability-Guided 3DGS Optimization
- Weight generated-views by hallucination, frame, and pixel reliability



Results

- **Contributions**
  - **1) Geometry-aware scene grounding**
    - Stabilizes sparse-view 3DGS geometry before video diffusion generation
  - **2) Scene-grounded video diffusion prior**
    - Preserves observed regions while synthesizing unseen regions
  - **3) Reliability-guided 3DGS optimization**
    - Uses frame- and pixel-level reliability to suppress hallucinated pseudo-views.

Quantitative Results

Comparison on Mip-NeRF360 and Tanks & Temples outdoor scenes with 9 input views.

Dataset	Scene	*Without Diffusion*						*With Diffusion*					
		Baseline			Our Baseline			GuidedVD			Ours		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Mip-NeRF360	Garden	17.51	0.378	0.438	18.58	0.422	0.445	18.95	0.425	0.437	19.00	0.450	0.419
	Bicycle	15.44	0.262	0.528	17.23	0.342	0.496	17.51	0.339	0.507	17.74	0.354	0.480
	Stump	16.78	0.287	0.532	17.67	0.342	0.524	17.89	0.354	0.530	17.98	0.370	0.518
	Flowers	12.62	0.195	0.629	13.13	0.218	0.613	13.69	0.224	0.596	13.72	0.227	0.591
Tanks & Temples	Truck	14.80	0.502	0.402	15.15	0.541	0.398	16.44	0.551	0.399	16.50	0.559	0.384
	Barn	14.14	0.457	0.530	14.44	0.464	0.511	14.40	0.474	0.516	14.45	0.479	0.507
	Train	13.78	0.478	0.456	13.95	0.504	0.458	14.28	0.492	0.470	14.32	0.508	0.453
	Caterpillar	12.73	0.312	0.558	13.28	0.348	0.564	13.53	0.354	0.554	13.59	0.355	0.550
	Ignatius	11.45	0.243	0.588	11.94	0.279	0.606	12.17	0.284	0.587	12.21	0.284	0.585

Qualitative Results